AS MATHEMATICS - MECHANICS 1 - DAY 5

Inclined planes

Resolving weight and friction

YOUR AIM TODAY

Resolve forces parallel and perpendicular to a slope.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

The ideas you need

Choose axes along and perpendicular to the plane so most forces sit on an axis.

- Weight mg acts vertically downward.
- Component down slope is mg sin theta.
- Component into slope is mg cos theta.
- If no other perpendicular force, R=mg cos theta.
- Friction opposes motion or impending motion along the plane.

MEMORY RULE

Parallel uses sin; perpendicular uses cos when theta is the plane angle.

Smooth plane

WORKED STEPS

Mass 3 kg on 30 degrees: acceleration down slope is g sin30=4.9 m/s^2.

Rough plane

WORKED STEPS

R=mg cos theta; find friction, then apply F=ma along slope.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Resolve weight of 5 kg parallel to a 30 degree slope.	
2. Resolve it perpendicular to the slope.	
3. Find R on a smooth 30 degree slope.	
4. Find acceleration down the smooth slope.	
5. If friction 10 N acts up the slope, find acceleration.	
6. For mu=0.2 at limiting equilibrium, write the equation along the slope.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	5g sin30=24.5 N down slope.
2	5g cos30=42.4 N into slope.
3	R=42.4 N.
4	a=g sin30=4.9 m/s^2.
5	5a=24.5-10, so a=2.9 m/s^2 down slope.
6	mg sin theta=mu mg cos theta, so tan theta=mu.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.